1. Diagonalize the matrix $\begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix}$

$\rightarrow A = \begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix}$

$\Rightarrow A - \lambda I = \begin{bmatrix} 2-\lambda & -1 \\ -3 & 4-\lambda \end{bmatrix}$

$\Rightarrow |A - \lambda I| = (2-\lambda)(4-\lambda) - 3$

$\Rightarrow 8 - 2\lambda - 4\lambda + \lambda^2 - 3$

$\Rightarrow \lambda^2 - 6\lambda + 5$

Now, $|A - \lambda I| = 0 \Rightarrow \lambda^2 - 6\lambda + 5 = 0$

$\Rightarrow \lambda^2 - 5\lambda - \lambda + 5 = 0$

$\Rightarrow \lambda(\lambda - 5) - 1(\lambda - 5) = 0$

$\therefore \lambda = 1, 5$

<u>When</u> $\lambda = 1$

Let, $v_1 = \begin{pmatrix} x \\ y \end{pmatrix}$

$Av_1 = \lambda v_1$

$\rightarrow \begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 2x + y \\ -3x + 4y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$

$$\Rightarrow \begin{cases} 2x+y = x \\ -3x+2y = y \end{cases}$$

$$\Rightarrow \begin{cases} x-y = 0 \\ -3x+3y = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x-y = 0 \\ x-y = 0 \end{cases}$$

1st and 2nd are Identical so,

$\therefore$ $x-y=0$

Here, $(2-1) = 1$ free variable, which is y.

$\therefore$ $y=1$, $x=1$

$\therefore$ $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

when,

$\lambda = 5$, $v_2 = \begin{pmatrix} x \\ y \end{pmatrix}$

we have,

$Av_2 = \lambda v_2$

$$\begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 5\begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 2x - y \\ -3x + 4y \end{bmatrix} = \begin{bmatrix} 5x \\ 5y \end{bmatrix}$$

$$\Rightarrow \begin{cases} 2x - y = 5x \\ -3x + 4y = 5y \end{cases}$$

$$\Rightarrow \begin{cases} -3x - y = 0 \\ -3x - y = 0 \end{cases}$$

$$= \begin{cases} 3x + y = 0 \\ 3x + y = 0 \end{cases}$$

1st and 2nd equations are identical

$\therefore$ $3x + y = 0$

Here, $(2-1) = 1$ free variable. which is y.

$x = -\frac{1}{3}$, $y = 1$

$\therefore$ $v_2 = \begin{pmatrix} -\frac{1}{3} \\ 1 \end{pmatrix}$

$$P = \begin{bmatrix} v_1 & v_2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -\frac{1}{3} \\ 1 & 1 \end{bmatrix}$$

$$\Rightarrow |P| = 1 + \frac{1}{3} = \frac{4}{3}$$

Now, $$P^{-1} = \frac{1}{\frac{4}{3}} \begin{bmatrix} 1 & \frac{1}{3} \\ -1 & 1 \end{bmatrix}$$

$$= \frac{3}{4} \begin{bmatrix} 1 & \frac{1}{3} \\ -1 & 1 \end{bmatrix}$$

$$\therefore D = P^{-1} A P$$

$$\Rightarrow \frac{3}{4} \begin{bmatrix} 1 & \frac{1}{3} \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{3} \\ 1 & 1 \end{bmatrix}$$

$$\Rightarrow \frac{3}{4} \begin{bmatrix} 1 & \frac{1}{3} \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{2}{3} - 1 \\ -3+4 & 1+4 \end{bmatrix}$$

$$\Rightarrow \frac{3}{4} \begin{bmatrix} 1 & \frac{1}{3} \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{5}{3} \\ 1 & 5 \end{bmatrix}$$

$$= \frac{3}{4} \begin{bmatrix} 1+\frac{1}{3} & -\frac{5}{3}+\frac{5}{3} \\ -1+1 & \frac{5}{3}+5 \end{bmatrix} = \frac{3}{4} \begin{bmatrix} \frac{4}{3} & 0 \\ 0 & \frac{20}{3} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$$

Ans

② solve The linear equation with The Help of Argument matrix method.

$$x_1 + x_2 - 2x_3 = -3$$

$$2x_1 + 5x_2 + 3x_2 = 11$$

$$-x_1 + 3x_2 + x_3 = 5$$

solⁿ:

$$\left[\begin{array}{ccc|c} 1 & 1 & -2 & -3 \\ 2 & 5 & 3 & 11 \\ -1 & 3 & 1 & 5 \end{array}\right]$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & -2 & -3 \\ 0 & 3 & 7 & 17 \\ -1 & 3 & 1 & 5 \end{array}\right] \quad R_2' = R_2 - 2R_1$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & -2 & -3 \\ 0 & 3 & 7 & 17 \\ 0 & 4 & -1 & 2 \end{array}\right] \quad R_3' = R_3 + R_1$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -2 & -3 \\ 0 & 1 & 7/3 & 17/3 \\ 0 & 4 & -1 & 2 \end{array}\right] \quad R_2' = \frac{1}{3} R_2$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & -2 & -3 \\ 0 & 1 & 7/3 & 17/3 \\ 0 & 0 & -\frac{31}{3} & -\frac{62}{3} \end{array}\right] \quad R_3' = R_3 - 4R_2$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & -2 & -3 \\ 0 & 1 & \frac{7}{3} & 17/3 \\ 0 & 0 & 1 & 2 \end{array}\right] \quad R_3' = \frac{R_3}{-\frac{31}{3}}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & -2 & -3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right] \quad R_2 = R_2 - \frac{7}{3} R_3$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right] \quad R_1 = R_1 + 2R_3$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right] \quad R_1 = R_1 - R_2$$

$\therefore \; x = 0, \; y = 1, \; z = 2$

3 Find the inverse of $A = \begin{bmatrix} 5 & 8 & 1 \\ 0 & 2 & 1 \\ 4 & 3 & -1 \end{bmatrix}$ with the help of argumented matrix method.

Soln.

$$\left[\begin{array}{ccc|ccc} 5 & 8 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 4 & 3 & -1 & 0 & 0 & 1 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{ccc|ccc} 1 & \frac{8}{5} & \frac{1}{5} & \frac{1}{5} & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 4 & 3 & -1 & 0 & 0 & 1 \end{array}\right] \quad R_1' = \frac{R_1}{5}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 8/5 & 1/5 & \frac{1}{5} & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & -\frac{17}{5} & -\frac{9}{5} & -\frac{4}{5} & 0 & 1 \end{array}\right] \quad R_3' = R_3 - 4R_1$$

$$= \left[\begin{array}{ccc|ccc} 1 & 8/5 & 1/5 & 1/5 & 0 & 0 \\ 0 & 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & -\frac{17}{5} & -\frac{9}{5} & -\frac{4}{5} & 0 & 1 \end{array}\right] \quad R_2' = \frac{R_2}{2}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 8/5 & \frac{1}{5} & 1/5 & 0 & 0 \\ 0 & 1 & 1/2 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{10} & -\frac{4}{5} & \frac{17}{10} & 1 \end{array}\right] \quad R_3' = R_3 + \frac{17}{5} R_2$$

$$= \left[\begin{array}{ccc|ccc} 1 & 8/5 & 1/5 & 1/5 & 0 & 0 \\ 0 & 1 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & 8 & -17 & -10 \end{array}\right] \quad R_3' = R_3 \times -10$$

$$= \left[\begin{array}{ccc|ccc} 1 & 8/5 & 1/5 & 1/5 & 0 & 0 \\ 0 & 1 & 0 & -4 & 9 & 5 \\ 0 & 0 & 1 & 8 & -17 & -10 \end{array}\right] \quad R_2' = R_2 - \frac{R_3}{2}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 8/5 & 0 & -7/5 & 17/5 & 2 \\ 0 & 1 & 0 & -4 & 9 & 5 \\ 0 & 0 & 1 & 8 & -17 & -10 \end{array}\right] \quad R_1' = R_1 - \frac{R_3}{5}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & -11 & -6 \\ 0 & 1 & 0 & -4 & 9 & 5 \\ 0 & 0 & 1 & 8 & -17 & -10 \end{array}\right] \quad R_1' = R_1 - \frac{8}{5} R_2$$

$$A^{-1} = \begin{bmatrix} 5 & -11 & -6 \\ -4 & 9 & 5 \\ 8 & -17 & -10 \end{bmatrix}$$

Ans

(4) Show That the matrix $\begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 4 \\ -5 & 5 & 6 \end{bmatrix}$

satisfies caley Hamilton Theorem and Hence compute A^{-1}.

→ **Solⁿ:** $A - \lambda I = \begin{bmatrix} 2-\lambda & 0 & 1 \\ -2 & 3-\lambda & 4 \\ -5 & 5 & 6-\lambda \end{bmatrix}$

Now, $|A - \lambda I| = 0$

Trace of matrix A is $2+3+6 = 11$

Diagonal minors $= -2+17+6 = 21$

$\therefore \lambda^3 - 11\lambda^2 + 21\lambda - |A| = 0$

$$|A| = 2(-2) + 0 + 1(-10+15)$$

$$\Rightarrow -4 + 5$$
$$= 1$$

$$\therefore \lambda^3 - 11\lambda^2 + 21\lambda - 1 = 0$$

Replace λ by A -

$$A^3 - 11A^2 + 21A - I = 0$$

Now,
$$A^2 = A \times A = \begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 4 \\ -5 & 5 & 6 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 4 \\ -5 & 5 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 4-5 & 5 & 2+6 \\ -4-6-20 & 9+20 & -2+12+24 \\ 10-10-30 & 45 & 51 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 5 & 8 \\ -30 & 29 & 34 \\ -30 & 45 & 51 \end{bmatrix}$$

$$A^3 = A^2 A = \begin{bmatrix} -1 & 5 & 8 \\ -30 & 29 & 34 \\ -30 & 45 & 51 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 4 \\ -5 & 5 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} -52 & 55 & 67 \\ -288 & 257 & 290 \\ -405 & 390 & 456 \end{bmatrix}$$

$\therefore A^3 - 11A^2 + 21A - I$

$$= \begin{bmatrix} -52 & 55 & 67 \\ -288 & 257 & 290 \\ -405 & 390 & 456 \end{bmatrix} - 11 \begin{bmatrix} -1 & 5 & 8 \\ -30 & 29 & 34 \\ -30 & 45 & 51 \end{bmatrix}$$

$$+ 21 \begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 4 \\ -5 & 5 & 6 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ The matrix satisfy caley Hamilton Theorem

Now,

$$A^3 - 11A^2 + 21A - I = 0$$

$$A^3A^{-1} - 11A^2 \cdot A^{-1} + 21AA^{-1} - IA^{-1} = 0$$

$$A^2 \cdot AA^{-1} - 11A \cdot AA^{-1} + 21AA^{-1} - A^{-1} = 0$$

$$\Rightarrow A^2I - 11AI + 21I - A^{-1} = 0$$

$$\Rightarrow A^{-1} = A^2 - 11A + 21$$

$$\Rightarrow \begin{bmatrix} -1 & 5 & 8 \\ -30 & 29 & 34 \\ -30 & 45 & 51 \end{bmatrix} - 11\begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 4 \\ -5 & 5 & 6 \end{bmatrix} + 2\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} -21 & 5 & -3 \\ -8 & -2 & -10 \\ -85 & -10 & -13 \end{bmatrix}$$

Ans

5. compute singular value Decomposition (SVD) for the given matrix. $\begin{bmatrix} 2 & -2 \\ 1 & 1 \end{bmatrix}$

$\Rightarrow A = \begin{bmatrix} 2 & -2 \\ 1 & 1 \end{bmatrix}$

$A^T = \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}$

$AA^T = \begin{bmatrix} 2 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix}$

Now, $AA^T - \lambda I = \begin{bmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{bmatrix}$

$|AA^T - \lambda I| = 0$

$\therefore (8-\lambda)(2-\lambda) = 0$

$\therefore \lambda = 2, 8$

$\therefore D = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix}$ $\therefore$ Singular values $S = \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}$

~~Now we have~~

$\xrightarrow{\text{i}}$

Now $AA^T u_1 = \lambda u_1$ where, $u_1 = \begin{bmatrix} x \\ y \end{bmatrix}$

$\therefore \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8x \\ 8y \end{bmatrix}$

$\Rightarrow \begin{cases} 8x = 8x \\ 2y = 8y \end{cases} \Rightarrow \begin{cases} 8x = 8x \\ y = 0 \end{cases}$

(11)

Let,

$\therefore x = -1$

$\therefore u_1 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \Rightarrow |u_1| = \sqrt{(-1)^2 + 0^2} = 1$

Again:

$u_2 = \begin{bmatrix} x \\ y \end{bmatrix}$

$AA^T u_2 = \lambda u_2 \Rightarrow \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$

$\Rightarrow \begin{cases} 6x = 0 \\ 2y = 2y \end{cases}$

$\Rightarrow \begin{cases} x = 0 \\ 2y = y \end{cases}$

$\therefore$ Let, $y = 1$

$\therefore u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow |u_2| = \sqrt{0^2 + 1^2} = 1$

$\therefore U = (u_1 \quad u_2) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

Now, for $A^T A$

$A^T A = \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ 1 & 1 \end{bmatrix}$

$= \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix}$

$A^T A - \lambda I = \begin{bmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{bmatrix}$

$\Rightarrow |A^T A - \lambda I| = 0$

$\Rightarrow (5-\lambda)(5-\lambda) = 9$

$\Rightarrow \lambda^2 - 10\lambda + 16 = 0$

$\Rightarrow \lambda^2 - 8\lambda - 2\lambda + 16 = 0$

$\Rightarrow (\lambda - 8)(\lambda - 2) = 0, \ \lambda = 8, 2$

$$\therefore D = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\therefore S = \sqrt{D} = \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}$$

$A^T A v_1 = \lambda v_1$, Here, $v_1 = \begin{bmatrix} x \\ y \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8x \\ 8y \end{bmatrix} \quad [\text{when } \lambda = 8]$$

$$\Rightarrow \begin{cases} 5x - 3y = 8x \\ -3x + 5y = 8y \end{cases}$$

$$\Rightarrow \begin{cases} -3x - 3y = 0 \\ -3y - 3y = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x + y = 0 \\ x + y = 0 \end{cases}$$

$\Rightarrow x + y = 0$

$(2-1) = 1$ free variable, which is y

$\therefore y = 1, \ x = -1 \quad \therefore v_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$|v_1| = \sqrt{(-1)^2+1^2} = \sqrt{2}$

$\therefore v_1 = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$

Again, $A^TAv_2 = \lambda v_2$, where $v_2 = \begin{bmatrix} x \\ y \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 2\begin{bmatrix} x \\ y \end{bmatrix}$ $[\lambda = 2]$

$\Rightarrow \begin{cases} 3x - 3y = 0 \\ -3x + 3y = 0 \end{cases}$

$\Rightarrow \begin{cases} x - y = 0 \\ x - y = 0 \end{cases}$

$\Rightarrow x - y = 0$

$(2-1) = 1$ free variable, which is y

$y = 1$, $x = 1$, $\therefore v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$|v_2| = \sqrt{1^2+1^2} = \sqrt{2}$

$\therefore v_2 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$

$\therefore V = \begin{bmatrix} -\sqrt{2}/2 & \sqrt{2}/2 \\ \frac{\sqrt{2}}{2} & \sqrt{2}/2 \end{bmatrix}$

$$V^T = \begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$\therefore A = USV^T$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -2\sqrt{2} \times \frac{\sqrt{2}}{2} & 2\sqrt{2} \times \frac{\sqrt{2}}{2} \\ \sqrt{2} \times \frac{\sqrt{2}}{2} & \cancel{0}\sqrt{2} \times \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -2 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cancel{0}2 & -2 \\ 1 & 1 \end{bmatrix}$$

Ans